

SANCHAY – Implementation Master Document

Full Product Blueprint: Architecture + Execution + Granular Tasks

1. System Architecture

Frontend (React) → FastAPI Backend → Core Engines → PostgreSQL + Redis

Core Engines

- 1 Portfolio Engine – handles transactions & valuation
- 2 Analytics Engine – computes XIRR, CAGR
- 3 AI Engine – rule + clustering + narrative
- 4 Reporting Engine – PDF generation

2. Epics → Features → Tasks

Epic 1: User & Authentication

- 1 Design user table schema
- 2 Implement JWT authentication
- 3 Create login/signup APIs
- 4 Add role-based access middleware

Epic 2: Client CRM

- 1 Create client table schema
- 2 Build client CRUD APIs
- 3 Implement risk profiling form
- 4 Store interaction logs
- 5 Enable advisory notes system

Epic 3: Portfolio Engine

- 1 Design portfolio schema
- 2 Create transaction model
- 3 Implement buy/sell logic
- 4 Build holdings calculator
- 5 Map assets to prices

Epic 4: Data Ingestion

- 1 Write EOD price scraper
- 2 Store daily prices
- 3 Schedule cron jobs
- 4 Validate data integrity

Epic 5: Analytics

- 1 Implement XIRR formula
- 2 Build CAGR calculator
- 3 Compute allocation %
- 4 Compare with benchmarks

Epic 6: AI Engine

- 1 Extract portfolio features
- 2 Implement rule-based alerts
- 3 Apply clustering (KMeans)
- 4 Build narrative generator templates

Epic 7: Reporting

- 1 Design report templates
- 2 Generate PDF reports
- 3 Include performance + gains
- 4 Schedule periodic reports

3. API Contract (Sample)

- 1 POST /auth/login
- 2 POST /clients/create
- 3 GET /clients/list
- 4 POST /portfolio/create
- 5 POST /transaction/add
- 6 GET /portfolio/performance
- 7 GET /reports/generate

4. Data Flow

User Input → API → Portfolio Engine → DB → Analytics Engine → AI Engine → Reporting → UI

5. AI Decision Flow

Portfolio Data → Feature Extraction → Rule Engine → Clustering → Narrative Output

6. Deployment Plan

- 1 Setup AWS EC2
- 2 Deploy FastAPI app
- 3 Configure PostgreSQL
- 4 Setup Redis
- 5 Enable cron jobs
- 6 Configure domain & HTTPS